Proficient in CATIA (80 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hours)**
1	Session 1	**Introduction to CATIA V5** What is CATIA V5 Design intent System requirements Getting started **Understanding CATIA interface** Workbenches Menus and toolbars Finding tools Specification tree Compass Graphic properties Message bar Document management **Introduction to sketcher** Entering sketcher workbench Basic sketching Positioned sketching **Profile creation** Profile Predefined profile Circle Spline Conic Line Point Axis **Understanding standard toolbar** New Open Save	Theory/Demo	2.0 Hrs
2	Session 2	Practise: Create sketches with profile tools	Practicals	2.0 Hrs
3	Session 3	**Understanding view manipulation** Fly mode Fit all In Pan Rotate Zoom in Zoom out Normal view Multi view Quick view View mode Hide/Show Swap visible space **Understanding standard toolbar** Save management Print Cut Copy Paste Undo Redo **Understanding sketch tools** **Creating construction/ standard element** **Applying geometrical/ dimensional constraints** **Understanding user selection filter toolbar** **Understanding visualization toolbar** Cut part by sketch plane 3D Element visualization D Element visualization Diagnostics Dimensional constraints Geometrical constraints	Theory/Demo	2.0 Hrs
4	Session 4	Practise: Creating sketches with profile tools	Practicals	2.0 Hrs
		Editing profile Corner Chamfer Relimitations Transformation		

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5	Session 5	3D geometry **Understanding constraints** Constraints Constraint creation Contact constraint Fix together Auto constraint Animate constraint Edit Multi – constraint **Sketch analysis**	Theory/Demo	2.0 Hrs
6	Session 6	Practise: Create fully constrained sketches (profile) using sketcher workbench	Practicals	2.0 Hrs
7	Session 7	**Introduction to part design** **Creating sketch-based features** Pad definition Drafted filleted pad definition Pocket definition Drafted filleted pocket Shaft definition Groove definition Hole definition **Creating reference element** Plane definition	Theory/Demo	2.0 Hrs
8	Session 8	Practise: Fully constrained sketches and sketch based features. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
9	Session 9	**Creating sketch-based feature** Rib definition Slot definition Stiffener definition Solid combine definition **Applying dress-up features** Edge fillet definition Variable fillet definition Chordal fillet definition Face-face fillet definition Tritangent fillet definition Chamfer definition	Theory/Demo	2.0 Hrs
10	Session 10	Practise: Create 3D models with features as discussed in session 9. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
11	Session 11	**Creating sketch-based features** Multi-sections solid definition Removed multi-section solid definition **Applying dress-up features** Draft definition Draft reflect line definition Variable angle draft definition Shell definition Thickness definition Thread/tap definition Remove/replace face definition	Theory/Demo	2.0 Hrs
12	Session 12	Practise: Create 3D models with features as discussed in session 11. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
13	Session 13	**Creating reference element** Point definition Line definition **Creating transformation features** Transformation definition Rotation definition Symmetry definition Axis to Axis definition Mirror Definition Pattern – Rectangular/ Circular pattern definition Scale definition **Applying boolean operations** Assemble definition Add definition Remove definition Intersect definition Union trim definition Remove lump definition	Theory/Demo	2.0 Hrs

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14	Session 14	Practise: Create 3D models using boolean features and application of transformation features. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
15	Session 15	**Applying advanced dress-up features** Draft both sides Advanced draft Automatic fillet Automatic draft **Understanding Tools toolbar** Update all Manual update mode Axis system Mean dimensions Create datum Only current body/ Only current operated solid Catalog browser **Understanding Measure Toolbar** Measure between Measure item Measure inertia	Theory/Demo	2.0 Hrs
16	Session 16	Practise: Application of draft features and tools discussed in session 15. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
17	Session 17	**Understanding knowledge toolbar** Formula Relations Knowledge inspector Lock Design table Equivalent dimensions **Apply Material** **Data Exchange in CATIA**	Theory/Demo	2.0 Hrs
18	Session 18	Practise: Create smart features using knowledge tools. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
19	Session 19	**Introduction to Assembly Design** Understanding top down & bottom up approach **Understanding Product Structure Tools** New component New product New part Existing component Existing component with positioning **Understanding Constraints Toolbar** Coincidence constraints Contact constraints Offset constraints Angle constraints Fix component Fix together Quick constraint Flexible/Rigid sub-assembly Change constraint Reuse pattern	Theory/Demo	2.0 Hrs
20	Session 20	Practise: Build assembly using top-down and bottom-up approach. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
21	Session 21	**Understanding Move Toolbar** Manipulation Snap/Smart move Explode Stop manipulate on clash **Understanding Product Structure Tools** Replace component Graph tree reordering Generate numbering Selective load Manage representations Multi instantiation **Understanding Constraint Creation Toolbar** Default mode Chain mode Stock mode **Assembly Features** Split	Theory/Demo	2.0 Hrs

		Hole		
		Pocket		
		Add		
		Remove		
		Symmetry		
		Creating publications		
		Applying material		
		Creating scenes		
		Enhanced scene		
		Scene browser		
		Applicative entities reordering		
		Understanding space analysis toolbar		
		Understanding catalogue browser		
22	Session 22	Practise: Practise application of assembly features and scenes. Practise workbook exercise and hands on exercise	Practicals	2.0 Hrs
23	Session 23	**Large assembly management**	Theory/Demo	2.0 Hrs
		Display settings in options dialog box		
		Screen settings		
24	Session 24	Practise: Large assembly management	Practicals	2.0 Hrs
25	Session 25	**Introduction to sheet metal design**	Theory/Demo	2.0 Hrs
		Understanding sheet metal parameters		
		Understanding views toolbar		
		Fold/Unfold		
		Multi viewer		
		Views management		
		Creating walls		
		Wall		
		Wall on edge		
		Extrusion		
		Swept walls- Flange, Hem, Tear Drop, User Flange		
		Creating rolled walls		
		Hopper		
		Free from surface		
		Rolled wall		
		Understanding bending operations		
		Bend / Conical bend		
		Bend from flat		
		Unfolding/ Folding		
		Point or curve mapping		
		Recognize definition		
26	Session 26	Practise: Construct sheet metal design using sheet metal features in CATIA V5	Practicals	2.0 Hrs
27	Session 27	**Understanding cutting operations**	Theory/Demo	2.0 Hrs
		Cut-out		
		Circular cut-out		
		Hole		
		Corner relief		
		Corner		
		Chamfer		
		Understanding Stamping Operations		
		Surface stamping		
		Bead		
		Curve stamping		
		Flanged cut out		
		Louver		
		Bridge		
		Flanged hole		
		Circular stamp		
		Stiffening rib		
		Dowel		
		User stamp		
		Understanding Transformation Operations		
		Mirror		
		Rectangular pattern		
		Circular pattern		
		User pattern		
		Translation		
		Rotation		
		Symmetry		
		Axis to axis		
		Understanding Manufacturing Preparation		
		Check overlapping		

		Save as DXF		
28	Session 28	Practise: Construct sheet metal design using sheet metal features in CATIA V5	Practicals	2.0 Hrs
29	Session 29	**Wireframe and Surface Design** **Introduction to Surface Design** **Understanding Wireframe Toolbar** Point, point and planes repetition Line, axis, polyline Plane, planes between Projection Intersection **Creating Surfaces** Extruded surface definition Revolution surface definition Sphere surface definition **Trim & Split Definition**	Theory/Demo	2.0 Hrs
30	Session 30	Practise: Construct surface models using wireframe and surface features in CATIA V5	Practicals	2.0 Hrs
31	Session 31	**Creating Surfaces** Cylinder surface definition Offset surface definition Fill surface definition Multi-Sections surface definition Blend definition **Understanding Operations Toolbar** Join Healing Untrim Disassemble **Creating Sweep Surface** Swept surface definition using profile type: - Explicit **Understanding Operations Toolbar** Extract Transformations commands Extrapolate	Theory/Demo	2.0 Hrs
32	Session 32	Practise: Construct surface models using wireframe and surface features in CATIA V5	Practicals	2.0 Hrs
33	Session 33	**Introduction to drafting** **Creating new drawing file** **Creating projection views** Front view Un folded view View from 3D Projection Auxiliary Isometric Advanced front view **Creating sectional views** Offset section view Aligned section view Offset section cut Aligned section cut	Theory/Demo	2.0 Hrs
34	Session 34	Practise: Create projection and section views of the 3D models created in earlier sections	Practicals	2.0 Hrs
35	Session 35	**Creating details views** Detail Sketched detail profile Quick detail Sketched quick detail profile **Creating clipping view** Clipping Sketched clipping profile Quick detail Sketched quick clipping profile **Creating broken view** **Creating broke out view** **Add 3D clipping** **Creating wizard** **Adding drawing sheets** Sheets New view Instantiate 2D component	Theory/Demo	2.0 Hrs

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		Deriving annotations		
		Generate dimension		
		Generate dimension step by step		
		Balloon creation		
		Bill of materials		
		Advanced bill of materials		
36	Session 36	Practise: Create detailed views and annotations of the 3D models	Practicals	2.0 Hrs
37	Session 37	**Applying dimensioning**	Theory/Demo	2.0 Hrs
		Creating dimensions using		
		Dimensions		
		Chained dimensions		
		Cumulated dimensions		
		Stacked dimensions		
		Length dimension		
		Angle dimension		
		Radius dimension		
		Diameter dimension		
		Chamfer dimension		
		Thread dimension		
		Co-ordinate dimension		
		Hole dimension table		
		Co-ordinate dimension table		
		Dimension edition using		
		Re-route dimension		
		Create interruption(s)		
		Remove interruption(s)		
		Create/Modify clipping		
		Remove clipping		
		Understanding tolerancing toolbar		
		Datum feature		
		Geometrical feature		
		Understanding annotations toolbar		
		Text		
		Symbols		
		Table		
		Add leader		
		Understanding dress up toolbar		
		Axis and thread		
		Area fill		
		Arrow		
		Understanding geometry creation toolbar		
		Points		
		Line		
		Circle and ellipse		
		Profiles		
		Curves		
		Understanding geometry modification toolbar		
		Relimitations		
		Transformation		
		Constraints		
38	Session 38	Practise: Create detailed views and annotations of the 3D models	Practicals	2.0 Hrs
39	Session 39	**Defining standards**	Theory/Demo	2.0 Hrs
		Inserting frame and title block		
		Understanding links		
		Understanding positioning toolbar		
		Element positioning		
		Line up		
		Align into system		
		Analyse		
		Show geometry in all viewpoints		
		Dimension analysis		
		Dimension generation		
		Filters		
		Analysis		
		Learning realtime rendering workbench		
40	Session 40	Practise: Create title block, BOM and practise rendering of 3D models using CATIA V5	Practicals	2.0 Hrs